

Event-Driven Alpha

Predicting Post-News Stock Volatility with Machine Learning

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Three Questions

1. Can machine learning rank which news events will be followed by the greatest remaining volatility?
2. Which information sources contribute most to that ranking?
3. Can the predictive signal produce profitable executable long-straddle trades?

Why Predict Movement Instead of Direction?

Direction

Will the stock go up or down?

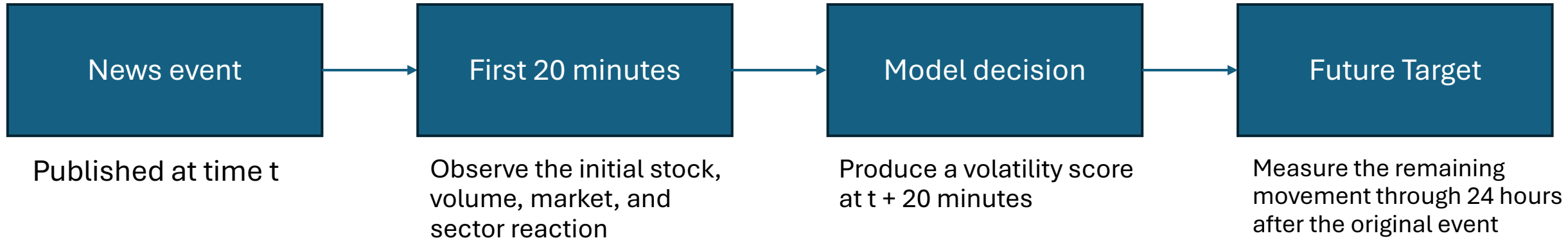
- Requires predicting the sign
- Depends heavily on prior expectations
- Positive-sounding news can still produce a decline
- Negative-sounding news can still produce a rally

Movement / volatility

How much will the stock move?

- Large rises and large declines both count
- Captures whether an event was market-moving
- Does not require predicting the correct sign
- Produces a natural event-ranking target

What the Model Predicts



Features: Information available by the decision time

Target: Realized variance from future five-minute returns

Data and Evaluation

Final matched cohort

- 134,584 events
- 571 tickers
- January 2016–June 2026

Information joined to each event

- News and event metadata
- Stock, market, sector, and volume data
- Historical option information
- Macroeconomic and SEC context

Evaluation

- Five expanding chronological folds
- 87,480 out-of-fold predictions
- Models train only on earlier events

How Ranking Performance Is Judged

Spearman correlation ↑

- Do higher model scores correspond to higher realized volatility?

Score-decile enrichment ↑

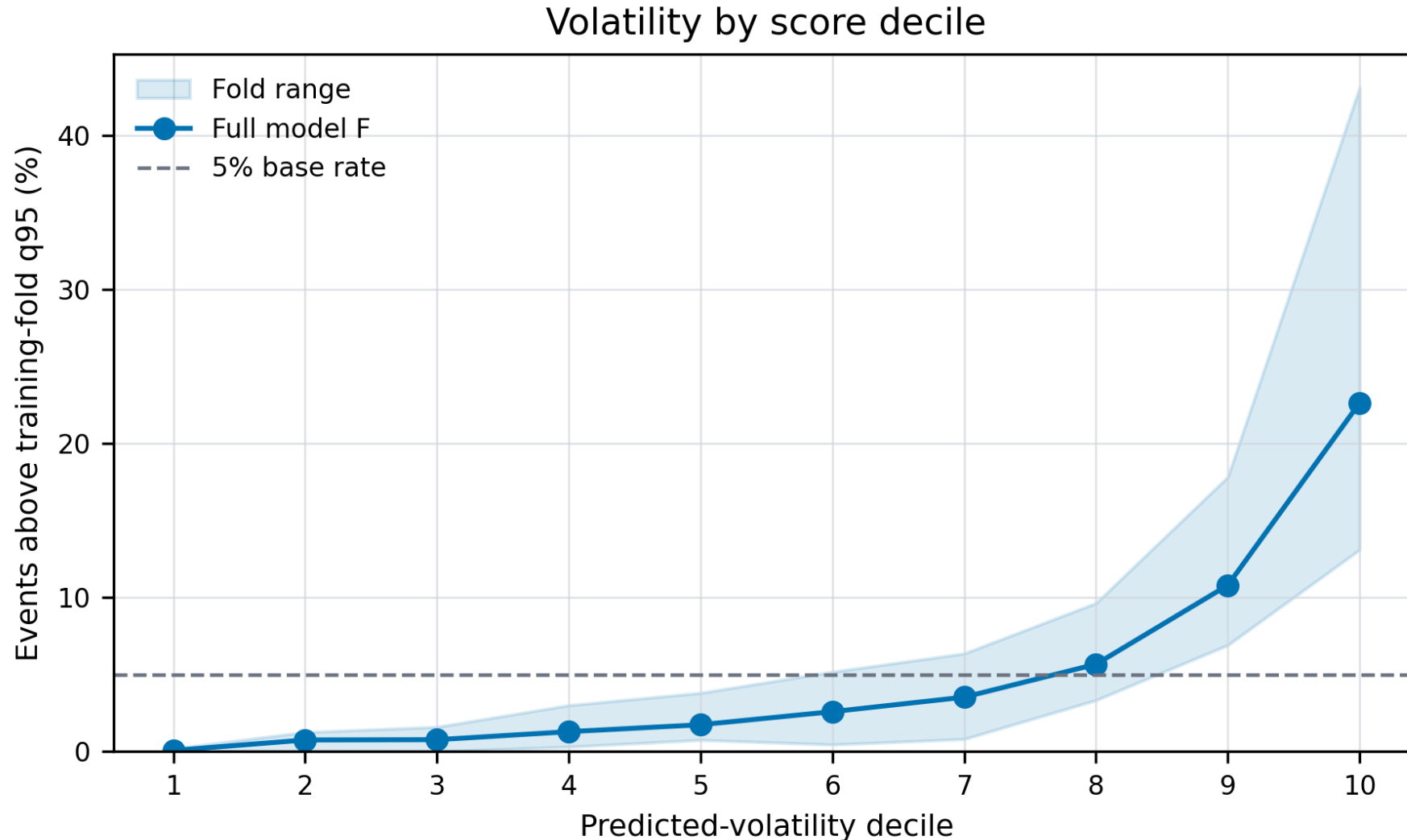
- Are extreme-volatility events concentrated among the highest scores?

QLIKE ↓

- Is the predicted variance scale reasonable?

Extreme event = realized variance above the corresponding training fold's 95th percentile

Question 1 Result: High Scores Identify Extreme Volatility



Question 1 Answer: The Ranking Works

0.626

Pooled out-of-fold Spearman

23% versus 5%

Top-decile extreme-event rate
versus base rate

1.243 versus 1.868

Full-model QLIKE versus lagged-
market QLIKE

0.0001

Fold-aware permutation p-value

The model meaningfully ranks remaining post-news
volatility on unseen chronological periods.

Question 2: Which Information Sources Improve the Ranking?

Same Ridge model and same events; only the available information changes

Cumulative sequence

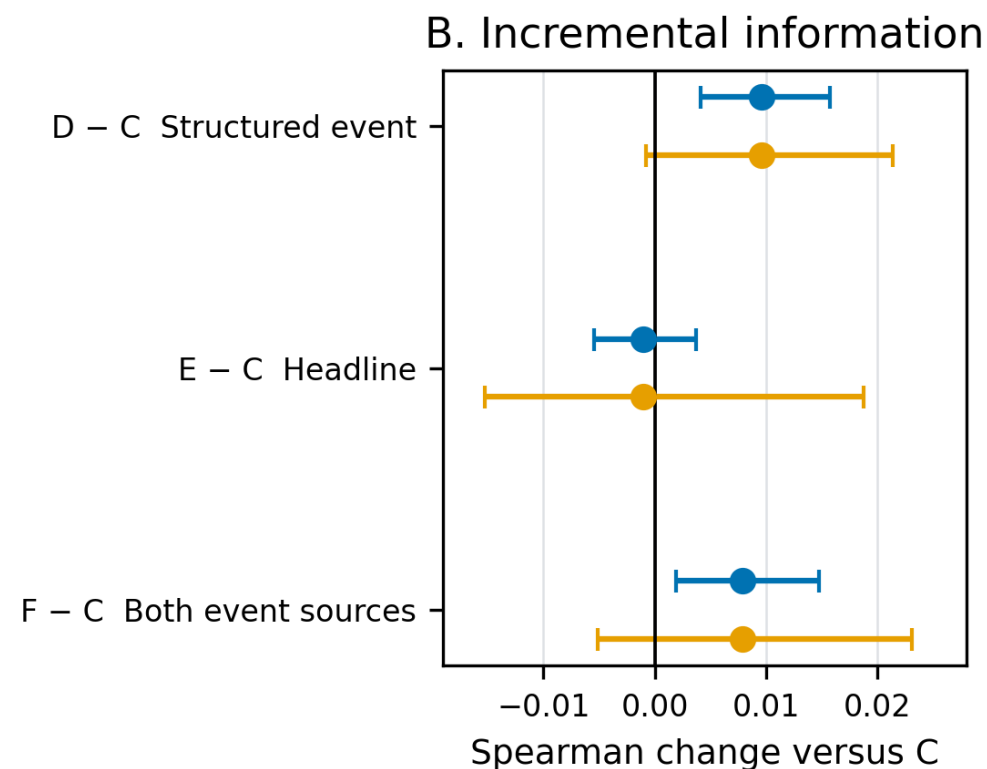
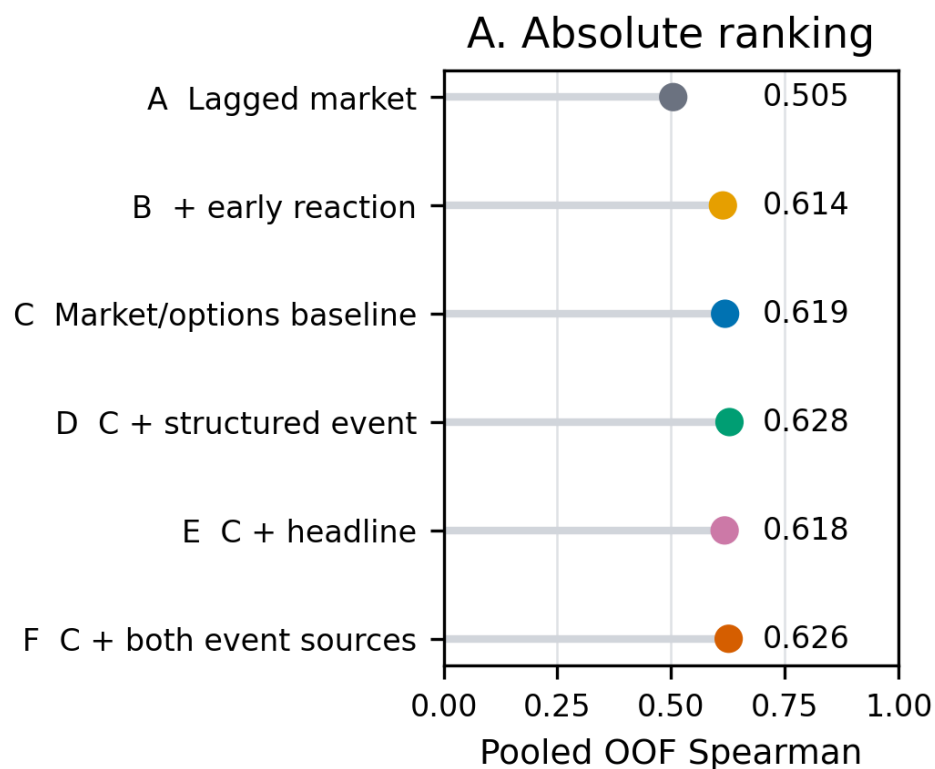
Variant	Information
A. Lagged market	Pre-event volatility, volume, event history, ticker, and time controls
B. Initial reaction	A plus the first 20 minutes of stock, market, sector, and volume response
C. Options and macro	B plus historical option and macroeconomic information

Separate extensions of C

Variant	Information added to C
D. Structured event	Source, event category, timestamp quality, and SEC context
E. Headline text	Headline TF-IDF
F. Both event sources	Structured information and headline text

Question 2 Result: Most of the Gain Comes from the Early Reaction

Signal sources



Question 2 Answer: Early Market Behavior Contains Most of the Signal

Initial market reaction

- Largest incremental gain: approximately **+0.109 Spearman**

Historical options and macro

- Smaller additional gain: approximately **+0.005 Spearman**

Structured event information

- Small positive estimate, but not stable across calendar months

Headline TF-IDF

- No improvement over the strong market, options, and macro baseline

News remains essential because it identifies the ticker and the time at which the prediction begins.

Question 3: Why Test a Long Straddle?

Call

Tends to gain value when the stock rises

+

Put

Tends to gain value when the stock falls

=

Long straddle

Buy a call and put at approximately the same strike

A straddle can benefit from a sufficiently large move in either direction.

Why Large Movement Can Still Lose Money

The option premium already prices expected movement

- Anticipated events can have expensive calls and puts
- A large realized move can still be smaller than the movement priced into the straddle

Executable prices are not the midpoint

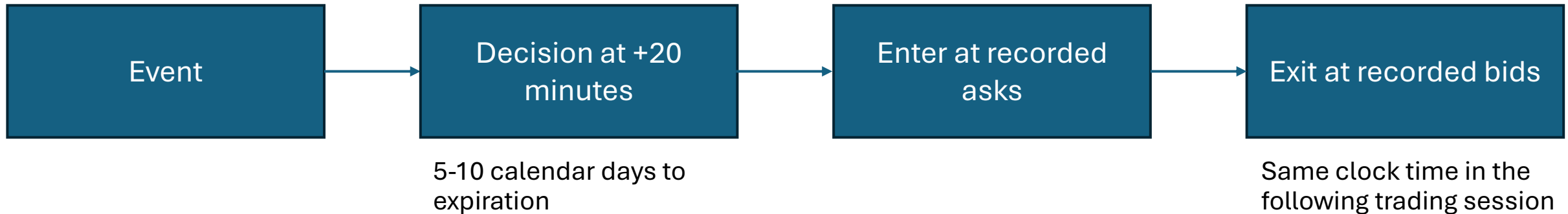
Bid	Midpoint	Ask
\$4.90	\$5.00	\$5.10

- Immediate buyer pays the ask
- Immediate seller receives the bid
- The difference is the bid/ask spread

Straddle P&L = call exit bid + put exit bid - call entry ask - put entry ask

Predefined Trading Policies and Execution

Policy	Model target
H2	Probability that the straddle produces positive net P&L
H3	Continuous return-related outcomes, ranked by predicted return on premium



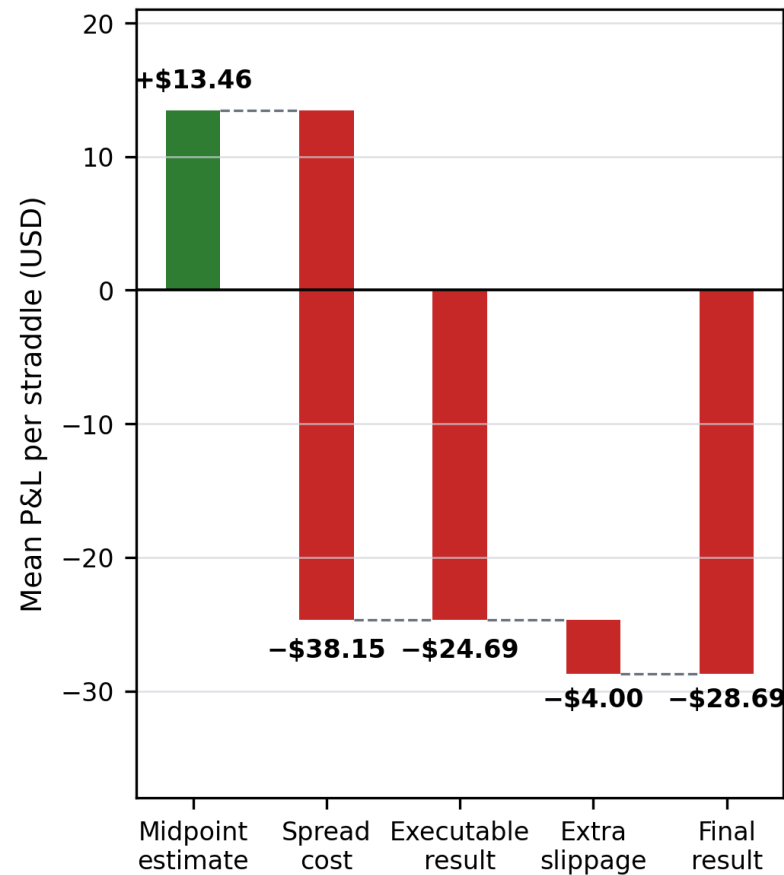
7,485 events with exact contract-level option data

Premium-weighted return = total net P&L ÷ total premium paid

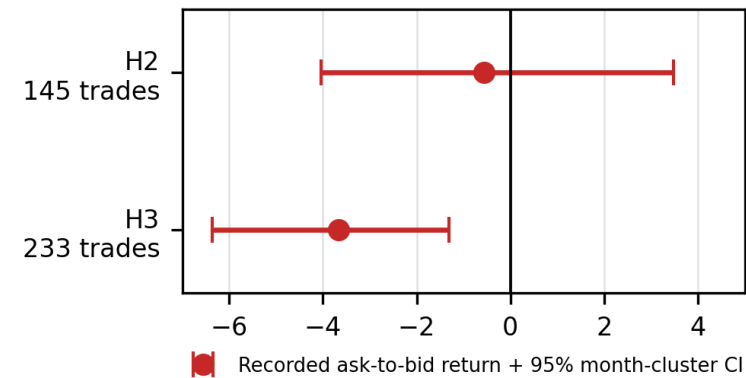
Question 3 Result: The Trading Policies Were Not Profitable

Prediction-to-trading gap

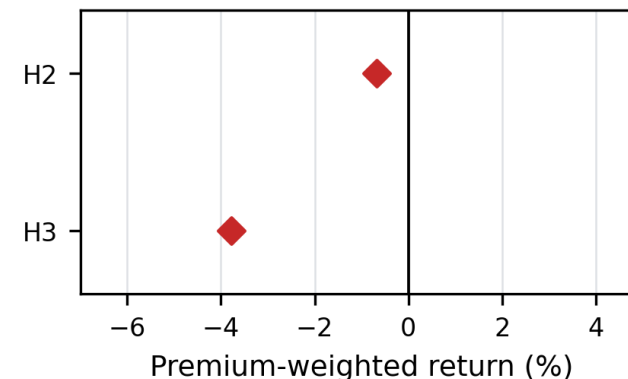
A. Execution-cost example across eligible development events



B. Predefined straddle policies



C. Slightly worse execution
One adverse tick per option leg



Question 3 Answer: The Model Finds Volatility, Not Underpricing

Volatility ranking asks

- Which events will move more than other events?

Profitable straddle trading asks

- Will the resulting option value exceed what was paid for the call and put?

- Option premiums already contain expected movement
- The first 20 minutes also give the option market time to reprice
- Buying at the ask and selling at the bid consumes additional value
- **A correct volatility ranking does not automatically identify a pricing error.**

Final Conclusions

1. Post-news volatility is predictable

- Pooled Spearman: **0.626**
- Highest decile: **23% extreme events**
- Base rate: **5%**

2. Early market behavior contains most of the incremental signal

- Initial reaction adds approximately **0.109 Spearman**
- Explicit event content adds limited robust improvement

3. The tested straddle policies are not profitable

- H2: **-0.558%**
- H3: **-3.661%**

Machine learning can rank post-news volatility, but the tested strategies do not identify volatility that is sufficiently underpriced to trade profitably.